

BALACHARIGE PAWAN MENUKA

Software Engineering Undergraduate

Colombo, Sri Lanka | 0716389275 | pawanmenuka02@gmail.com
linkedin.com/in/pawan-menuka-1356593aa | github.com/Pawan-Menuka

SUMMARY

Third-year Software Engineering undergraduate who designs and ships production full-stack systems: a deployed AI speech-coaching platform, a dual-market e-commerce platform with secure token-based authentication and idempotent payment webhooks, and a concurrency-safe appointment booking microservice. Comfortable across TypeScript/Next.js frontends, Node.js and Express REST APIs, PostgreSQL and MongoDB data modelling, and containerised deployment with CI/CD. Seeking a software engineering internship in full-stack or backend development.

EDUCATION

BSc (Hons) in Information Technology, Specialising in Software Engineering

Sri Lanka Institute of Information Technology (SLIIT), Malabe, Sri Lanka

Expected graduation: 2027

- CGPA: 3.69 / 4.00
- Dean's List: Year 1 Semester 1, Year 2 Semester 2

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, SQL, Solidity, HTML, CSS

Frontend: React, Next.js (App Router, SSR/SSG/ISR), React Native (Expo), Tailwind CSS, Recharts

Backend: Node.js, Express, REST API design, JWT authentication, role-based access control, Prisma, Mongoose, Zod

Databases: PostgreSQL, MongoDB, Redis, Supabase

Cloud and DevOps: Docker, Kubernetes, GitHub Actions CI/CD, Vercel, Cloudflare R2, Neon, Railway

Messaging and Background Jobs: RabbitMQ, BullMQ

Testing: Vitest, Jest, Foundry (fuzz and invariant testing), Slither

Tools: Git, GitHub, Postman, Jira, Swagger/OpenAPI, pnpm monorepos, Figma

Concepts: Microservices, event-driven architecture, concurrency control, API security, Agile/Scrum

PROJECTS

IMPROMPTU SPEECH TRAINER — AI Speech Coaching Platform (Solo)

Next.js 16, React 19, TypeScript, PostgreSQL, Prisma, Clerk, AssemblyAI, Claude API, Cloudflare R2, Upstash Redis, Vercel

Live: impromptu.pawanmenuka.com | GitHub: github.com/Pawan-Menuka/Impromptu-Speech

- Built and deployed a full-stack web application that delivers timed impromptu speaking practice and returns per-criterion AI feedback, with scoring rubrics that tighten as difficulty increases.
- Engineered the record-to-score pipeline end to end: browser audio capture, object storage on Cloudflare R2, AssemblyAI transcription with filler-word detection and words-per-minute analysis, and LLM-based scoring against difficulty-scaled rubrics.
- Implemented a provider fallback from the Claude API to Gemini after a live third-party outage during development, keeping the scoring path available under upstream failure.
- Added distributed rate limiting on Upstash Redis to bound per-user AI spend, and modelled sessions and scores in PostgreSQL with Prisma to power a Recharts progress dashboard.
- Deployed on Vercel with GitHub Actions CI and a Cloudflare R2 CDN serving audio and optimised image assets.

ARANYA CEYLON — Full-Stack E-Commerce Platform (Solo)

Next.js 14 (App Router), TypeScript, Express, Prisma, PostgreSQL, Stripe, PayHere, Vitest

GitHub: github.com/Pawan-Menuka/Aranya-Ceylon

- Architected a pnpm monorepo comprising a Next.js storefront, an Express REST API and a shared Zod schema package, giving end-to-end TypeScript type safety between client and server.
- Implemented authentication with short-lived JWTs, opaque refresh-token rotation and token-family reuse detection, routing all browser traffic through a same-origin backend-for-frontend proxy so tokens remain in HttpOnly cookies.

- Designed a dual-market commerce engine (local LKR and international USD) with market-scoped pricing and separate payment gateways, including idempotent Stripe webhook handling and atomic checkout and refund transactions with stock restoration.
- Modelled the catalog in PostgreSQL with Prisma across 14 migrations, using full-text search, cursor pagination and integer-cents money arithmetic to eliminate floating-point pricing errors.
- Developed a role-gated admin console covering revenue reporting, inventory and image management, order fulfilment and refunds, and an append-only audit log; backed by a Vitest regression suite and GitHub Actions CI.

SMART HEALTHCARE PLATFORM — Appointment Management Microservice (Group Project)

Node.js, Express, MongoDB, Redis, BullMQ, RabbitMQ, JWT, Docker, Kubernetes, React

Role: Service owner for the appointment domain | GitHub: github.com/Vihanga-Deemantha/Smart-Health-Care

- Designed and built the appointment microservice end to end within a microservices healthcare platform, including its integrations with the doctor, patient, payment, notification and telemedicine services.
- Engineered a concurrency-safe two-phase hold-then-book flow using Redis slot holds with TTL expiry, per-patient hold limits, MongoDB multi-document transactions and a unique compound index that prevents double-booking under race conditions.
- Implemented the full appointment lifecycle (hold, booked, confirmed, rescheduled, completed, cancelled, no-show) with a dual-party attendance confirmation model and a policy-driven cancellation cutoff.
- Built an event-driven integration layer over RabbitMQ and BullMQ background workers for automatic hold release, 24-hour and 1-hour reminders and waitlist auto-promotion; containerised the service with Docker, authored Kubernetes manifests and documented the API with OpenAPI/Swagger.

DECENTRALIZED FREELANCE ESCROW PROTOCOL (Solo)

Solidity, Hardhat, Foundry, OpenZeppelin, Chainlink, The Graph, React, wagmi/viem

GitHub: github.com/Pawan-Menuka/Decentralized-Escrow-Platform

- Designed and implemented milestone-based escrow smart contracts with a per-milestone state machine, pull-payment withdrawals, milestone-level dispute handling and OpenZeppelin ReentrancyGuard and Pausable patterns; deployed and verified on the Sepolia testnet.
- Validated contract invariants with Foundry fuzz and invariant test suites and Slither static analysis in GitHub Actions CI, and integrated Chainlink Price Feeds for USD-denominated milestone pricing and Chainlink Automation for time-locked auto-release.

LEADERSHIP AND ACTIVITIES

AIESEC — iGV Value Delivery (Front Office) and Project Audits

January 2024 — January 2025

- Worked in the iGV (Incoming Global Volunteer) front office on the Value Delivery team, supporting incoming exchange participants through onboarding, project matching and in-country delivery, and escalating issues to the relevant project leads.
- Managed project audits for the entire function over a one-year term, reviewing documentation and compliance for member-led projects across multiple teams and working with project leads to close gaps before sign-off.
- Maintained the function's shared cloud drive as the central store for project records — organised the folder hierarchy, managed access permissions across teams, and kept naming and versioning consistent so documentation stayed retrievable through leadership handovers.

CERTIFICATIONS

- AWS Certified Developer — Associate (DVA-C02) — in progress, exam scheduled October 2026

REFERENCES

Mr. Eranga Bowatte — Business Partner — HR Executive, Virtusa (Pvt) Ltd.

ebowatte@virtusa.com | +94 76 590 0816

Mr. Nelum Amarasena — Senior Lecturer, Dept. of Information Technology, SLIIT

nelum.a@sliit.lk | +94 71 909 6100